ModelIt!

Teacher's Guide

Engineering a Solution: The Design Process

Modeling How Iteration Turns Failure into Innovation

Quick Reference

Grade Level	Grade 06L2
NGSS Standard	MS-ETS1-1, MS-ETS1-2
Time	45–60 minutes (core lesson)
Materials	Student Activity Pack, devices with ModelIt! access
Prep	Print activity packs, test ModelIt! on student devices
Grouping	Partners or small groups (2–4)

ModelIt! — Discovery Collective

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1 NGSS Standards & CAST Alignment

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Performance Expectation: MS-ETS1-1, MS-ETS1-2

Define the criteria and constraints of a design problem with sufficient precision to ensure a successful solution, taking into account relevant scientific principles and potential impacts on people and the natural environment. Evaluate competing design solutions using a systematic process to determine how well they meet the criteria and constraints of the problem.

Three-Dimensional Learning

Science Practice	**Designing Solutions** — Students model the engineering design process showing how problem complexity, research, design quality, prototype performance, and iteration interact as a system.
Disciplinary Core Idea	**ETS1.A Defining and Delimiting Engineering Problems** — The more precisely a problem is defined in terms of criteria and constraints, the more likely it is that the designed solution will be successful. Specifications and constraints help ensure that a solution addresses the intended problem.
Crosscutting Concept	**Influence of Engineering, Technology, and Science on Society** — Students explore how the iterative design process has produced innovations that improve human life, and how engineering solutions must consider societal impact and environmental constraints.

CAST Testing Alignment

MULTIPLE CHOICE:

Two engineering teams are given the same problem. Team A spends 80% of their time researching and 20% building one prototype. Team B spends 20% researching and 80% building and testing three prototypes. Based on the engineering design process model, which team is more likely to produce a better solution, and why?

CONSTRUCTED RESPONSE:

Using your engineering design process model, explain why SpaceX's approach of rapidly testing, failing, and iterating on rocket designs led to reusable rockets, while other companies that spent years perfecting designs on paper before testing fell behind. Reference the feedback loop between prototype performance and design quality.

2 Scoring Rubric

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Assessment Reference

This is the same rubric students see in their Activity Pack. Use it during and after the lesson for formative and summative assessment. Circle or highlight the proficiency level for each skill.

Skill	Expert (4)	Proficient (3)	Developing (2)	Beginning (1)
Identifying Components	EXPERT Correctly identifies and sorts ALL components as internal or external with detailed reasoning	PROFICIENT Correctly identifies and sorts MOST components with some reasoning	DEVELOPING Identifies components but has difficulty sorting or explaining	BEGINNING Needs significant help identifying components
Establishing Relationships	EXPERT Accurately describes ALL positive and negative relationships with clear evidence	PROFICIENT Accurately describes MOST relationships with some evidence	DEVELOPING Identifies some relationships but struggles with direction	BEGINNING Needs help understanding how components connect
Running Simulations	EXPERT Makes accurate predictions AND explains surprising results using evidence	PROFICIENT Makes reasonable predictions and records clear observations	DEVELOPING Runs simulations but predictions or observations are incomplete	BEGINNING Needs help running simulations or recording data
Explaining Results	EXPERT Uses multiple pieces of evidence to explain HOW and WHY systems interact	PROFICIENT Uses evidence to explain system interactions	DEVELOPING Gives partial explanations with limited evidence	BEGINNING Needs help connecting evidence to explanations

Scoring Guidance

Expert (4): Student demonstrates deep understanding and can transfer knowledge to new contexts.
Proficient (3): Student meets grade-level expectations and shows solid understanding.
Developing (2): Student is making progress but needs additional scaffolding or practice.
Beginning (1): Student needs significant support; consider using Differentiation strategies (Section 7).

3 Background Content Knowledge

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For Your Confidence

This section provides the science background you need to facilitate this lesson with authority. You don't need to teach all of this — it's here so you can answer student questions accurately.

The engineering design process is not a simple linear path from problem to solution. It is a dynamic, iterative cycle where testing, failure, and redesign are the most important steps. Understanding this process reveals why innovation is messy, why failure is valuable, and why the best solutions come from persistent iteration.

Define, Research, Design, Test, Iterate

The engineering design process has five major stages: defining the problem (criteria and constraints), researching existing solutions and relevant science, designing a potential solution, building and testing a prototype, and iterating (using test results to improve the design). The process is cyclical, not linear: each test sends you back to redesign with new knowledge.

Why First Designs Fail

Real-world problems involve countless interacting variables. No amount of planning can predict every behavior of a physical system. Wind load, material fatigue, user behavior, manufacturing tolerances, and environmental conditions all create surprises. The ONLY way to discover these problems is to test. Every failure reveals information that was impossible to obtain otherwise.

The Power of Iteration

Research on engineering outcomes consistently shows that iterative designs outperform single-attempt designs. SpaceX crashed multiple rockets before achieving controlled landings. The Wright Brothers tested over 200 wing shapes in a wind tunnel. Apple creates dozens of prototypes for each product. The feedback loop (test, learn, redesign, retest) is the engine of engineering innovation.

Criteria, Constraints, and Trade-offs

Every engineering problem has criteria (what the solution MUST do) and constraints (limitations like budget, materials, time, safety, and environmental impact). Solutions must satisfy criteria while respecting constraints, which often requires trade-offs. Making a bridge stronger might require more material (cost) or more weight. Engineering judgment is about making the best trade-offs.

4 The LEVER Framework

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About LEVER

LEVER is how every ModelIt! lesson works. Students move through five phases to build, test, and refine a computational model. Each lesson is self-contained — students complete all five phases in one session.

L **Locate the System**
Students identify problem complexity and research depth as the two external inputs, and design quality, prototype performance, and iteration count as the internal responses of the engineering design system.

E **Establish Relationships**
Students determine that problem complexity negatively affects initial design quality, research depth positively affects design quality, design quality positively affects prototype performance, and iteration count positively improves prototype performance through a feedback loop.

V **Visualize & Model**
Students build a model showing how research reduces the negative impact of complexity, and how the iteration feedback loop (test, learn, redesign) progressively improves performance.

E **Evaluate Outcomes**
Students run no-research-rush vs. research-first scenarios, then observe how adding iterations dramatically improves outcomes even with lower initial quality.

R **Revise & Extend**
Students add budget constraint or time pressure to explore how real-world limitations affect the engineering process and force trade-offs.

Dropping ModelIt! Into Your Existing Unit

ModelIt! is a **supplement**, not a replacement for your science curriculum. Each lesson stands alone, but it becomes even more powerful when students revisit their model throughout a unit. Here are easy ways to weave it in:

At the start of a unit	Use this lesson to introduce the phenomenon. Students build a "first draft" model based on what they already know. Save it.
After a reading or lab	Have students reopen their model and update it: "Based on what you just learned, would you add a component? Change a relationship?" Takes 10–15 minutes.
As a "what-if" tool	When students ask questions during class, say: "Great question — go test that in your model!" Students run new scenarios and report back.
For peer discussion	Partners compare models: "Why does your model have this arrow and mine doesn't?" Differences spark scientific debate using evidence.
As a presentation tool	Students project their model and walk the class through it live — showing components, running scenarios, answering audience "what-if" challenges in real time.
At the end of a unit	Compare Day 1 model to final model. The growth IS the assessment — students see how their thinking evolved.

5 Slide-by-Slide Facilitation

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Slide 1: Cover

2 min

Say: *"SpaceX crashed SIX rockets before they landed one. Was that failure? Or was that the process working exactly as designed?"*

Watch For

Show a rapid montage of SpaceX rocket crashes followed by the first successful landing. Let students react.

Slide 2: Learning Objectives

3 min

Say: *"Today we are going to model the engineering design process itself and discover why failure is the most important step."*

Watch For

Have students read objectives. Pre-teach 'criteria,' 'constraints,' and 'iteration.'

Slide 3: Big Question

3 min

Say: *"Every product you own went through dozens of failures before it worked. Why can't smart engineers get it right the first time?"*

Watch For

Think-pair-share: Have you ever built something that did not work the first time? What did you do?

Slide 4: LEVER Framework

2 min

Say: *"Our model has a FEEDBACK LOOP. Testing tells you what is wrong, which makes the next design better, which reveals new problems to fix."*

Watch For

Draw the feedback loop on the board. This is different from the one-way models we have built before.

Slide 5: Activity 1: Components

8 min

Say: *"What can we control? What responds? And where is the feedback loop hidden?"*

Watch For

Guide sorting. Discuss why problem complexity is external (given) and research depth is external (chosen by the engineer).

Slide 6: Activity 2: Connections

8 min

Say: *"Here is the most important arrow in the entire model: prototype testing feeds BACK to design quality. This loop is what makes engineering work."*

Watch For

Trace the feedback loop: design -> prototype -> test results -> better design -> better prototype. Each cycle improves.

Slide 7: Activity 3: Simulation

10 min

Say: *"Let's race: a team that does no research vs. a team that researches first vs. a team that iterates three times. Who wins?"*

Watch For

Students predict outcomes. Run all three scenarios. The iterative team wins even with less initial research.

Slide 8: Discoveries

5 min

Say: *"Iteration beats everything. You can overcome a bad start by testing and improving. But you cannot overcome skipping the testing loop."*

Watch For

Connect to real examples: Wright Brothers, SpaceX, Apple, medical devices. Every great innovation went through painful iteration.

Slide 9: STEM Challenge

12 min

Say: *"Three drops. Three chances. After each drop, you analyze, redesign, and try again. The team that improves the MOST wins."*

Watch For

Set up the egg drop challenge with clear criteria and material constraints. Emphasize documentation of changes between iterations.

6 Answer Key & Expected Responses

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Component Sorting

- **EXTERNAL:** Problem Complexity, Research Depth
- **INTERNAL:** Design Quality, Prototype Performance, Iteration Count

Relationships

- **Problem Complexity to Design Quality:** NEGATIVE (-) — More complex problems have more interacting constraints, making it harder to produce a good first design. Complexity increases the chance of unexpected problems.
- **Research Depth to Design Quality:** POSITIVE (+) — More research means better understanding of the problem, existing solutions, and relevant science. This leads to better-informed designs that address more criteria from the start.
- **Design Quality to Prototype Performance:** POSITIVE (+) — A better design on paper translates to a better-performing prototype. Good planning reduces (but never eliminates) unexpected failures during testing.
- **Iteration Count to Prototype Performance:** POSITIVE (+) — Each test-improve cycle catches problems and implements solutions. More iterations mean more opportunities to identify and fix issues, progressively improving performance.
- **Prototype Performance to Design Quality:** POSITIVE (+) FEEDBACK LOOP — Testing reveals problems that were invisible on paper. This information directly improves the next design iteration. The loop creates progressive improvement with each cycle.

Key Discoveries

- Harder problems produce worse first designs, but research can partially offset complexity
- Research depth has a strong positive effect on initial design quality because understanding the problem is half the solution
- Iteration is the most powerful improvement tool. Each test-improve cycle catches problems the previous one missed
- The feedback loop between prototype testing and design quality is POSITIVE: testing informs better designs, which produce better prototypes, which reveal subtler problems
- Even expert engineers rarely succeed on the first try. Iteration is not failure; it IS the engineering process

Multiple Choice Post-Assessment Answer Key

1. In the engineering design model, what happens when a team jumps into building a prototype for a complex problem without doing any research first?

Correct: B) Design quality is low because the team does not understand the problem well enough, leading to poor prototype performance

Correct! The model shows that research depth has a strong positive effect on initial design quality. Without understanding the problem, existing solutions, and relevant science, the first design addresses the wrong issues or misses key constraints. This leads to a poor-performing prototype that wastes the first iteration.

2. Why is iteration the single most powerful variable for improving engineering outcomes?

Correct: B) Each test-improve cycle reveals problems that planning alone cannot predict, and fixing those problems creates a positive feedback loop of improvement

Correct! Real-world testing reveals unexpected problems that no amount of planning could anticipate. Each iteration identifies specific failures, which inform targeted improvements. This creates a positive feedback loop: testing reveals problems, redesign fixes them, the next test reveals subtler problems, and so on. Each cycle produces a measurably better design.

3. The model shows a positive feedback loop between prototype testing and design quality. What does this mean?

Correct: B) Testing a prototype provides specific information about what works and what fails, which directly improves the next design, which produces an even better prototype

Correct! In this positive feedback loop, testing reveals real performance data and failure points. This information improves the next design iteration. The improved design produces a better prototype, which reveals subtler (previously hidden) problems when tested. Each cycle drives improvement because real data continuously feeds back into the design process.

4. Two teams tackle the same engineering problem. Team A researches thoroughly and builds one prototype. Team B does minimal research but builds and tests three prototypes. Based on the model, which team likely produces a better solution?

Correct: B) Team B, because three iterations of testing and improvement typically outperform one well-researched attempt that has not been tested against reality

Correct! While research improves the first design, the model shows that iteration is the most powerful improvement variable. Three test-improve cycles reveal real-world problems that research alone cannot predict. Team B's iterative approach, even with less research, produces more total improvement through the feedback loop between testing and redesign.

5. How does increasing problem complexity affect the engineering design process?

Correct: B) More complex problems produce worse first designs because more constraints and criteria interact in unpredictable ways, making research and iteration even more important

Correct! Higher problem complexity means more criteria, more constraints, and more interactions between them. This makes it harder to design a good solution on the first try because there are more ways things can go wrong. The model shows that both research (to understand the complexity) and iteration (to discover unexpected interactions) become more critical as complexity increases.

7 Collaborative Learning Protocols

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Why This Matters

ModelIt! is a digital tool, but the deepest learning happens between humans. Computational modeling is most powerful when students build understanding *together* — debating relationships, challenging predictions, and constructing explanations as a community. These protocols ensure every student's voice is part of the model. Use at least ONE per lesson.

Model Gallery Walk

When: After students build models (LEVER Phase V)
Setup: Groups post their models on screens/desks. Half the class stays to "defend," half rotates with sticky notes.
Protocol: Visitors leave 2 sticky notes: one "I notice..." and one "I wonder..." After rotation, defenders read feedback aloud. Class discusses patterns.
Why it works: Students see that the SAME phenomenon can be modeled differently — and that's how real science works. (Bandura: learning through observation of peers)

Scientist Fishbowl

When: During simulation discussion (LEVER Phase E)
Setup: 4-5 students sit in center circle. Rest of class observes with clipboards.
Protocol: Inner circle discusses: "What happened in Scenario 3 and WHY?" using only evidence from their models. Outer circle records strongest arguments. Swap groups.
Why it works: Students practice scientific discourse without teacher directing. Observers learn to evaluate arguments. (Freire: dialogue as the foundation of critical consciousness)

Component Jigsaw

When: During component analysis (LEVER Phase L)
Setup: Assign each group ONE component to become "experts" on. They research it, discuss it, draw it.
Protocol: Regroup so each new team has one expert per component. Experts teach their component to the team. Team builds the model together using everyone's expertise.
Why it works: Every student is essential — the model literally cannot be built without each person's knowledge. Interdependence drives engagement. (Lave & Wenger: learning through legitimate peripheral participation)

Prediction Debate

When: Before running simulations (LEVER Phase E)
Setup: Post a scenario. Students physically move to corners: "Fire Spread will INCREASE a lot" / "INCREASE a little" / "STAY the same" / "DECREASE."
Protocol: Each corner group has 2 minutes to build their argument using model logic. One spokesperson presents. Then RUN the simulation. Losing corners explain what they missed.
Why it works: Movement + social risk + immediate feedback. Students CARE about the outcome because they committed publicly. Wrong predictions become the best learning moments.

Pair Programming for Models (Digital Collaboration)

When: Throughout model building
Setup: Two students, one device. **Driver** controls the mouse/keyboard. **Navigator** directs strategy and catches errors. Switch every 5 minutes.
Protocol: Navigator must verbalize every decision: "I think we should add a negative arrow from Rainfall to Dry Vegetation because..." Driver cannot act without Navigator's reasoning.
Why it works: Forces articulation of scientific thinking. The model becomes a shared artifact, not an individual assignment. Students who struggle with writing can demonstrate understanding verbally. Borrowed from computer science — where the best code is written in pairs.

Quick Protocol Selection Guide

Short on time (5 min)	Prediction Debate — gets bodies moving, generates energy fast
Full period available	Component Jigsaw — deep collaboration, every student contributes

After simulation day	Model Gallery Walk — peer feedback, multiple perspectives
Advanced discussion	Scientist Fishbowl — builds academic discourse skills
Every day default	Pair Programming — transforms individual work into dialogue

8 Differentiation & Extensions

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Below Grade Level

- Pre-sort components as a whole class
- Provide sentence frames for relationships
- Use visual vocabulary cards
- Pair with a strong reader

Advanced Learners

- Add 2-3 additional components
- Research real-world data
- Create multi-step causal chains
- Compare to similar systems globally

English Language Learners

- Pre-teach key vocabulary
- Provide bilingual glossary
- Use visual supports with text
- Allow verbal responses

Gifted & Talented

- Create feedback loop diagrams
- Research prevention strategies
- Write a policy brief
- Connect to climate change

9 Common Misconceptions

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Why This Matters

Addressing misconceptions proactively helps students build accurate scientific understanding. Use these as formative assessment checkpoints during the lesson.

STUDENTS MAY THINK	SCIENTIFIC UNDERSTANDING
"Good engineers get it right the first time"	The best engineers expect failure on the first attempt. They design EXPERIMENTS, not final products, in early iterations. NASA, SpaceX, Apple, and every major engineering organization budgets for multiple prototypes. Getting it right the first time is not a sign of good engineering; it usually means the problem was too simple.
STUDENTS MAY THINK	**SCIENTIFIC UNDERSTANDING**
"Failure means the design is bad"	In engineering, failure is DATA. Every failure tells you specifically what does not work and why. A prototype that survives a test without revealing any problems is actually LESS useful than one that fails, because the failure provides actionable information for improvement. The only real failure is not learning from the test.
STUDENTS MAY THINK	**SCIENTIFIC UNDERSTANDING**
"More research always beats more testing"	Research provides theoretical understanding, but testing provides real-world data that research cannot. The ideal approach combines both: research to build a strong foundation, then iterate to refine. Research alone misses problems that only emerge in physical testing. Testing alone wastes time on problems research could have prevented.